

## 02\_Graph\_Theory\_Graph 图-板子

图的存储:

[B3643 图的存储 - 洛谷](#)

```
#include<bits/stdc++.h>
// #define int long long
using namespace std;
using ll=long long;

void work(){
    int n,m;
    cin>>n>>m;          vector<set<int> > g(n+1);
    vector<vector<int> > adj(n+1,vector<int>(n+1,0));
        for(int i=1;i<=m;i++){
            int x,y;
            cin>>x>>y;
            g[x].insert(y);
            g[y].insert(x);
            adj[x][y]=1;
            adj[y][x]=1;
        }
        for(int i=1;i<=n;i++){
            for(int j=1;j<=n;j++){
                cout<<adj[i][j]<<" ";
            }
            cout<<"\n";
        }
        for(int i=1;i<=n;i++){
            cout<<g[i].size()<<" ";
            for(auto y:g[i]){
                cout<<y<<" ";
            }
            cout<<"\n";
        }
}

signed main(){
    ios::sync_with_stdio(0);
    cin.tie(0);
    int t=1;
    // cin>>t;
    while(t--){
        work();
    }
}

/*
5 5
1 2
2 3
3 5
1 3
3 4
*/
```

```

0 1 1 0 0
1 0 1 0 0
1 1 0 1 1
0 0 1 0 0
0 0 1 0 0
2 2 3
2 1 3
4 1 2 4 5
1 3
1 3
*/

```

图的遍历:

[B3862 图的遍历 \(简单版\) - 洛谷](#)

```

#include<bits/stdc++.h>
using namespace std;
using ll=long long;

void work(){
    int n,m;
    cin>>n>>m;          vector<set<int> > g(n+1);
    vector<vector<int> > adj(n+1,vector<int>(n+1,0));
    for(int i=1;i<=m;i++){
        int x,y;
        cin>>x>>y;
        g[x].insert(y);
    }

    int ans=0;
    vector<int> vis(n+1);
    auto dfs=[&](auto &&self,int x) -> void{
        ans=max(ans,x);
        for(auto y:g[x]){
            if(vis[y]) continue;
            vis[y]=1;
            self(self,y);
        }
    };
    for(int i=1;i<=n;i++){
        fill(vis.begin(),vis.end(),0);
        ans=0;
        dfs(dfs,i);
        cout<<ans<<' ';
    }
} signed main(){
    ios::sync_with_stdio(0);
    cin.tie(0);
    int t=1;
    // cin>>t;
    while(t--){
        work();
    }
}

```

